THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · HARD LEVEL CHAPTER TEST SET 1

Chapter 13: **Our Environment** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Assertion (A): The number of trophic levels in a food chain is limited. Reason (R): About 90 per cent of energy is lost at each trophic level. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

2. A food chain has four trophic levels. If the energy at the fourth level is 5 J, the energy at the first level is 1

(a) 50 J **(b)** 500 J **(c)** 50,000 J **(d)** 5000 J

3. Ozone in the stratosphere is formed by the action of 1

(a) CFCs on oxygen **(b)** UV radiation on oxygen
(c) visible light on nitrogen **(d)** infrared on ozone

4. Assertion (A): Decomposers are called the cleaners of the environment. Reason (R): They break down dead organisms into simple substances. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

5. If the energy available at the producer level is 4000 kJ, the energy transferred to the third trophic level is 1

(a) 40 kJ **(b)** 4 kJ **(c)** 400 kJ **(d)** 4000 kJ

6. A student claims that eating at a lower trophic level gives more energy. This is because 1

(a) food at lower levels is tastier **(b)** lower levels have more energy
(c) plants are free **(d)** the Sun is nearer

7. Assertion (A): Energy flow in an ecosystem is cyclic. Reason (R): Energy is lost as heat at every trophic level and cannot be reused by the organisms. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

8. In a food chain: plants, insects, frogs, snakes, eagles, the primary and the quaternary consumers are 1

(a) frogs and snakes **(b)** insects and eagles
(c) snakes and eagles **(d)** plants and insects

9. **Assertion (A):** Decomposers maintain the nutrient cycle in an ecosystem. 1
Reason (R): They break down dead organic matter into simple inorganic substances that return to the soil.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** The use of chlorofluorocarbons (CFCs) has damaged the ozone layer. 1
Reason (R): The ozone layer absorbs the harmful ultraviolet radiation from the Sun.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. In a food chain, grass, grasshopper, frog and snake, the energy in the frog is 50 J. Find the energy in the grass and in the snake. 2

12. Why is it not possible to have a food chain with 10 trophic levels? 2

13. In what way is a crop field different from a forest with respect to the diversity and the self-sustaining nature? 2

14. If the grass in a field has 8000 J of energy, find the energy in the tertiary consumer in the chain grass, rabbit, fox, wolf. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. In a pond food chain with plankton, small fish, big fish and kingfisher, the concentration of a pesticide in the plankton is 0.02 ppm. If it rises 10 times at each level, find the concentration in the kingfisher. 3

16. Suggest three activities that a school can take up to promote the 3 R's and explain how they help. 3

17. Why can the fish in a river downstream of a factory die? 3

OR

Explain why a rise in the population of one species may affect many others in a food web. 3

18. How does the burning of plastic waste affect the ecosystem? 3

OR

In an experiment, a beaker of vegetable waste and a beaker of plastic scraps are buried in soil. After a month what will be found? Explain. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** In a pond ecosystem, phytoplankton are eaten by small fish, small fish are eaten by big fish and big fish are eaten by fish-eating birds. The phytoplankton in the pond capture 10,000 J of solar energy. 5
(i) Name the producer in this food chain.
(ii) At which trophic level is the big fish?
(iii) How much energy is available to the big fish?
(iv) How much energy is available to the bird and why are food chains usually short?

20. Explain why the biomass of organisms decreases at each higher trophic level in a food chain. Give two consequences for the numbers of top carnivores in a forest. 5

OR

Describe in terms of trophic levels and energy why 1 hectare of land can feed more people if it is used to grow cereals than if it is used to rear cattle. Use numbers with the 10 per cent law. 5

END OF QUESTION PAPER — Total: 40 Marks